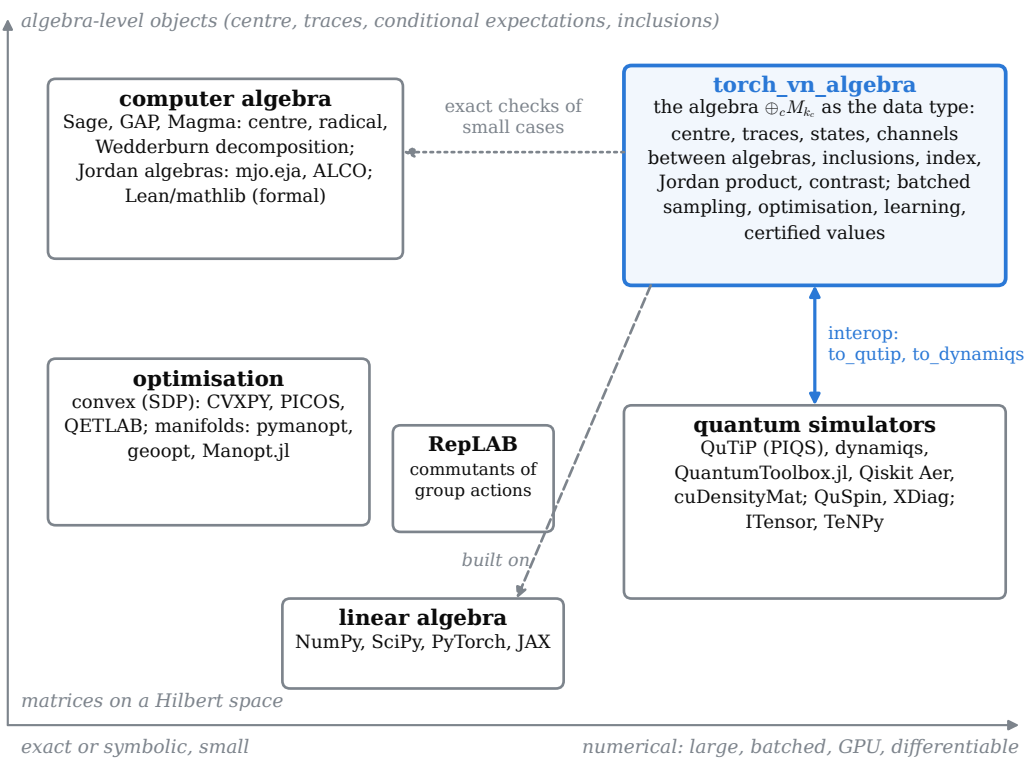
algebra-level objects (centre, traces, conditional expectations, inclusions)
computer algebra
Sage, GAP, Magma: centre, radical, Wedderburn decomposition; Jordan algebras: mjo.eja, ALCO; Lean/mathlib (formal)
exact checks of small cases
torch_vn_algebra
the algebra $\oplus_c M_{k_c}$ as the data type: centre, traces, states, channels between algebras, inclusions, index, Jordan product, contrast; batched sampling, optimisation, learning, certified values
interop: to_qutip, to_dynamiqs
optimisation
convex (SDP): CVXPY, PICOS, QETLAB; manifolds: pymanopt, geoopt, Manopt.jl
RepLAB
commutants of group actions
quantum simulators
QuTiP (PIQS), dynamiqs, QuantumToolbox.jl, Qiskit Aer, cuDensityMat; QuSpin, XDiag; ITensor, TeNPy
built on
linear algebra
NumPy, SciPy, PyTorch, JAX
matrices on a Hilbert space
exact or symbolic, small
numerical: large, batched, GPU, differentiable